

Global well-posedness of non-isentropic Navier-Stokes equation with BV initial data

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Outline

- 1 Introduction
- 2 Heat kernel
- 3 Local well-posedness of weak solution
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Model

1-D compressible Navier-Stokes in Lagrangian coordinate reads

$$\begin{cases} v_t - u_x = 0, \\ u_t + p_x = \left(\frac{\mu u_x}{v} \right)_x, \\ \theta_t + \frac{p}{c_v} u_x - \frac{\mu}{c_v v} (u_x)^2 = \left(\frac{\kappa}{c_v v} \theta_x \right)_x. \end{cases} \quad (1.1)$$

v : specific volume,

u : velocity,

θ : temperature,

$p(v, \theta) = \frac{\kappa \theta}{v}$: pressure,

μ, c_v, κ : constants.

Classical solution

J. Nash (1958,1962) and N. Itaya (1971,1976): Euler coordinates, local solution with regular initial data.

Y. I. Kanel (1968): Lagrangian coordinates, global solution with regular initial data.

A. Matsumura and T. Nishida (1980): Euler coordinates, small perturbation and global solution with regular initial data.

S. Kawashima (1987): General quasi-linear equations and obtain global solution for regular initial data.

Iteration scheme around constant state

Consider the isentropic equation

$$\begin{cases} v_t - u_x = 0, \\ u_t + p_x = \left(\frac{\mu u_x}{v} \right)_x, \end{cases}$$

Classical iteration scheme around constant state $(v^c, 0)$:

$$\begin{cases} v_t^{n+1} - u_x^{n+1} = 0, \\ u_t^{n+1} + (p(v^n))_x = \left(\mu \frac{u_x^{n+1}}{v^c} \right)_x + \left(\mu \frac{u_x^n}{v^n} - \mu \frac{u_x^n}{v^c} \right)_x. \end{cases}$$

Energy estimate requires higher order regularity due to the second order error terms.

Weak solution with discontinuity

D. Hoff (1991,1995): Full NS equation in Lagrangian coordinates with discontinuous initial data.

K. Chen, R. Hu, Q. Nguyen (2022): Local well-posedness with finite jumps.

S. Yu and T. Liu (2022): Global well-posedness of isentropic Navier-Stokes equation with BV initial data.

H. Wang and Z. (2023): Regularity and uniqueness of isentropic Navier-Stokes equation with BV initial data.

H. Wang, S. Yu and Z. (2022): Global well-posedness of non-isentropic Navier-Stokes equation with BV initial data.

R-H condition

Hyperbolic-parabolic equation with **R-H** condition:

$$\begin{aligned}s[v] &= [u], \\ s[u] &= [-p + \frac{\mu u_x}{v}].\end{aligned}$$

u is parabolic, thus $[u] = 0$; v is hyperbolic, thus $[v] \neq 0$:

- (i) $s = 0$: discontinuity will not move.
- (ii) $[-p + \frac{\mu u_x}{v}] = 0$: flux is continuous.

Weak solution

- ① (v, u) satisfy the equation in distribution sense for any test function $\varphi(x, t) \in C_0^\infty(\mathbb{R} \times [0, +\infty))$,

$$\begin{cases} \int_0^{+\infty} \int_{\mathbb{R}} \varphi (v_t - u_x) dx dt = 0, \\ \int_0^{+\infty} \int_{\mathbb{R}} \left[\varphi_x \left(\frac{\mu u_x}{v} - p \right) - \varphi_t u \right] dx dt = \int_{\mathbb{R}} \varphi(x, 0) u(x, 0) dx. \end{cases}$$

- ② The flux of u given by $\frac{\mu u_x}{v} - p$ is continuous with respect to x .

Iteration scheme around constant state fails

$$\begin{cases} v_t^{n+1} - u_x^{n+1} = 0, \\ u_t^{n+1} + (p(v^n))_x = \left(\mu \frac{u_x^{n+1}}{v^c} \right)_x + \left(\mu \frac{u_x^n}{v^n} - \mu \frac{u_x^n}{v^c} \right)_x. \end{cases}$$

Representation of the solutions:

$$v^{n+1}(x, t) = v_0(x) + \int_0^t u_x^{n+1}(x, s) ds$$

$$u^{n+1}(x, t) = \int_{\mathbb{R}} H(x - y, t) u_0(y) dy + \int_0^t \int_{\mathbb{R}} H_y(x - y, t) \mathcal{N} dy ds.$$

Idea

Iteration scheme when discontinuity occurs (Liu-Yu-2022):

$$\begin{cases} v_t^{n+1} - u_x^{n+1} = 0, \\ u_t^{n+1} + (p(v^n))_x = \left(\mu \frac{u_x^{n+1}}{v^n} \right)_x. \end{cases}$$

Key: **heat equation with discontinuous coefficient**
– Short-time Green's function.

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Heat kernel

$$\begin{cases} (\partial_t - \partial_x \rho(x, t) \partial_x) H(x, t; y, t_0; \rho) = 0, & t > t_0, \\ H(x, t_0; y, t_0; \rho) = \delta(x - y). \end{cases}$$

$\rho(x, t)$ is BV and satisfies

$$\begin{cases} \|\rho(\cdot) - C\|_{L^1} \leq \delta^*, & \|\rho(\cdot, t)\|_{BV} \leq \delta^*, \\ \|\rho_t(\cdot, t)\|_{\infty} \leq \delta^* \max\left(\frac{1}{\sqrt{t}}, 1\right), & 0 < \delta^* \ll 1, \\ \mathcal{D} := \{z \mid \rho(z, t) \text{ is not continuous at } z\}. \end{cases}$$

Note the discontinuity set $\mathcal{D}$ of $\rho(x, t)$ is independent of t .

Simple example: constant ρ

Case 1: If $\rho(x, t) = \kappa$ is a positive constant, then by Laplace transform we have

$$s\mathcal{L}[H](x, s; y) - \delta(x - y) = \kappa\mathcal{L}[H]_{xx}(x, s; y),$$

where we omit t_0 for simplicity. Solve the equation for $x > y$ and $x < y$ respectively, then we have

$$\mathcal{L}[H](x, s; y) = \frac{1}{2\sqrt{s\kappa}} e^{-\sqrt{\frac{s}{\kappa}}(x-y)}, \quad x > y, \quad \text{wave from } y \text{ to } x$$

$$\mathcal{L}[H](x, s; y) = \frac{1}{2\sqrt{s\kappa}} e^{\sqrt{\frac{s}{\kappa}}(x-y)}, \quad x < y, \quad \text{wave from } y \text{ to } x$$

$$H(x, t; y, t_0) = \frac{1}{\sqrt{4\pi\kappa(t-t_0)}} e^{-\frac{(x-y)^2}{4\kappa(t-t_0)}}, \quad \text{Inverse Transform}$$

Simple example: one discontinuity

Case 2: If $\rho(x, t) = \kappa_+ h(x) + \kappa_- (1 - h(x))$, $h(x)$ is Heaviside function. Then for $x < 0$:

$$\begin{aligned} \mathcal{L}[H](x, s; y) &= \frac{1}{2\sqrt{s\kappa_-}} e^{-\sqrt{\frac{s}{\kappa_-}}|x-y|} (1 - h(y)) \\ &+ \frac{1}{2\sqrt{s\kappa_-}} e^{-\sqrt{\frac{s}{\kappa_-}}|0-y|} R_{--} e^{\sqrt{\frac{s}{\kappa_-}}x} (1 - h(y)) \\ &+ \frac{1}{2\sqrt{s\kappa_-}} e^{-\sqrt{\frac{s}{\kappa_-}}|0-y|} T_{+-} e^{\sqrt{\frac{s}{\kappa_+}}x} h(y). \end{aligned}$$

For $x > 0$, we have

$$\begin{aligned}\mathcal{L}[H](x, s; y) &= \frac{1}{2\sqrt{s\kappa_-}} e^{-\sqrt{\frac{s}{\kappa_+}}|x-y|} h(y) \\ &\quad + \frac{1}{2\sqrt{s\kappa_+}} e^{-\sqrt{\frac{s}{\kappa_+}}|0-y|} R_{++} e^{-\sqrt{\frac{s}{\kappa_+}}x} h(y) \\ &\quad + \frac{1}{2\sqrt{s\kappa_+}} e^{-\sqrt{\frac{s}{\kappa_+}}|0-y|} T_{-+} e^{\sqrt{\frac{s}{\kappa_-}}x} (1 - h(y)),\end{aligned}$$

By R-H condition:

$$0 = [H] = [\rho(x, t)H_x],$$

We let x tend to 0 from both sides and derive

$$\left\{ \begin{array}{l} R_{--} = \frac{\kappa_- - \kappa_+}{\kappa_- + \kappa_+}, \\ T_{-+} = 1 + \frac{\kappa_- - \kappa_+}{\kappa_- + \kappa_+}, \\ R_{++} = \frac{\kappa_+ - \kappa_-}{\kappa_- + \kappa_+}, \\ T_{+-} = 1 + \frac{\kappa_+ - \kappa_-}{\kappa_- + \kappa_+}. \end{array} \right.$$

ρ is step function

Case 3: if $\rho(x, t) = \mu(x)$ is a step function satisfying the following properties,

$$\left\{ \begin{array}{l} \mathcal{D} \equiv \{\alpha \mid [\mu](\alpha) \neq 0\}, \\ \mu(z) = \kappa_i, \quad z \in (\alpha_i, \alpha_{i+1}), \quad i \in \mathbb{Z}, \\ \lim_{i \rightarrow \pm\infty} \alpha_i = \pm\infty, \quad \inf\{|\alpha_{i+1} - \alpha_i|, i \in \mathbb{Z}\} > \delta_*, \\ \|\mu\|_{BV} = \sum_{\alpha \in \mathcal{D}} |[\mu](\alpha)|. \end{array} \right.$$

The heat kernel as a path integral:

$$H(x, t; u, s) = \sum_{\omega \in \Omega_{y \rightarrow x}} \mathcal{A}[\omega] \frac{e^{-\frac{\mathcal{P}[\omega]^2}{4t}}}{\sqrt{4\pi t}},$$

where

$\Omega_{y \rightarrow x}$: all paths from y to x ,

$\mathcal{A}[\omega]$: reflection and transmission at discontinuities on path ω ,

$$\mathcal{P}[\omega] = \int_{\omega} \frac{dz}{\sqrt{\mu(z)}}.$$

Liu-Yu-2022

Suppose the **coefficient is a step function** and satisfies small condition as before, then we have

$$|H(x, t; y, t_0)| \leq C_* \frac{e^{-\frac{(x-y)^2}{C_*(t-t_0)}}}{\sqrt{t-t_0}},$$

$$|H_x(x, t; y, t_0)| + |H_y(x, t; y, t_0)| \leq C_* \frac{e^{-\frac{(x-y)^2}{C_*(t-t_0)}}}{t-t_0},$$

$$|H_{xy}(x, t; y, t_0)| + |H_t(x, t; y, t_0)| \leq C_* \frac{e^{-\frac{(x-y)^2}{C_*(t-t_0)}}}{(t-t_0)^{\frac{3}{2}}},$$

...

Liu-Yu-2022

Suppose the coefficient is BV and satisfies previous small condition, then for $t_0 < t < t_0 + t_\sharp$,

$$|\bar{H}(x, t; y, t_0; \rho)| \leq C_* \frac{e^{-\frac{(x-y)^2}{C_*(t-t_0)}}}{\sqrt{t-t_0}},$$

$$|\bar{H}_x(x, t; y, t_0; \rho)| + |\bar{H}_y(x, t; y, t_0; \rho)| \leq C_* \frac{e^{-\frac{(x-y)^2}{C_*(t-t_0)}}}{t-t_0},$$

$$|\bar{H}_{xy}(x, t; y, t_0; \rho)| + |\bar{H}_t(x, t; y, t_0; \rho)| \leq C_* \frac{e^{-\frac{(x-y)^2}{C_*(t-t_0)}}}{(t-t_0)^{\frac{3}{2}}},$$

...

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Iteration scheme

Perturbation around $(1, 0, 1)$:

$$\begin{cases} V_t^{n+1} - U_x^{n+1} = 0 \\ U_t^{n+1} - \left(\frac{\mu U_x^{n+1}}{1 + V^n} \right)_x = -p(1 + V^n, 1 + \Theta^n)_x \\ \Theta_t^{n+1} - \left(\frac{\kappa \Theta_x^{n+1}}{c_v(1 + V^n)} \right)_x = -\frac{p(1 + V^n, 1 + \Theta^n)}{c_v} U_x^n + \frac{\mu}{c_v(1 + V^n)} (U_x^n)^2, \\ (V^{n+1}, U^{n+1}, \Theta^{n+1})|_{t=0} = (v_0^*, u_0^*, \theta_0^*), \\ (V^0, U^0, \Theta^0) = (0, 0, 0). \end{cases} \quad (3.2)$$

Suppose $p = \frac{K\theta}{v}$ and the perturbation is small:

$$\|v_0^*\|_{L^1}, \|v_0^*\|_{BV}, \|u_0^*\|_{L^1}, \|u_0^*\|_{BV}, \|\theta_0^*\|_{L^1}, \|\theta_0^*\|_{BV} \leq \delta,$$

A priori estimates

Then,

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$$\left\{ \begin{array}{l} 0 < \delta, t_{\#} \ll 1, \quad 1 \leq n \leq k, \quad 0 < t < t_{\#}, \\ \max \left\{ \|U^n(\cdot, t)\|_{L_x^1}, \|U^n(\cdot, t)\|_{L_x^\infty}, \|U_x^n(\cdot, t)\|_{L_x^1}, \sqrt{t} \|U_x^n(\cdot, t)\|_{L_x^\infty} \right\} \leq 2C_{\#} \delta, \\ \max \left\{ \|\Theta^n(\cdot, t)\|_{L_x^1}, \|\Theta^n(\cdot, t)\|_{L_x^\infty}, \|\Theta_x^n(\cdot, t)\|_{L_x^1}, \sqrt{t} \|\Theta_x^n(\cdot, t)\|_{L_x^\infty} \right\} \leq 2C_{\#} \delta, \\ \max \left\{ \|V^n(\cdot, t)\|_{BV}, \|V^n(\cdot, t)\|_{L_x^1}, \|V^n(\cdot, t)\|_{L_x^\infty}, \sqrt{t} \|V_t^n(\cdot, t)\|_{L_x^\infty} \right\} \leq 2C_{\#} \delta, \\ \|V^n(\cdot, t) - V^n(\cdot, s)\|_{BV} \leq 2C_{\#} \delta \frac{(t-s)|\log(t-s)|}{\sqrt{t}}, \quad 0 \leq s < t, \\ \left| V^n(\cdot, t) \Big|_{x=z^-}^{x=z^+} \right| \leq 2 \left| v_0^*(\cdot) \Big|_{x=z^-}^{x=z^+} \right|, \quad z \in \mathcal{D}. \end{array} \right.$$

Local existence

Wang-Yu-Z.-2022

Suppose the initial data $(v_0^*, u_0^*, \theta_0^*)$ are small. Then there exists a weak solution

$$\left\{ \begin{array}{l} \delta > 0, \quad 0 < t < t_{\#} \ll 1, \\ \max \left\{ \|u(\cdot, t)\|_{L_x^1}, \|u(\cdot, t)\|_{L_x^\infty}, \|u_x(\cdot, t)\|_{L_x^1}, \sqrt{t} \|u_x(\cdot, t)\|_{L_x^\infty} \right\} \leq 2C_{\#} \delta, \\ \max \left\{ \|\theta(\cdot, t) - 1\|_{L_x^1}, \|\theta(\cdot, t) - 1\|_{L_x^\infty}, \|\theta_x(\cdot, t)\|_{L_x^1}, \sqrt{t} \|\theta_x(\cdot, t)\|_{L_x^\infty} \right\} \leq 2C_{\#} \delta, \\ \max \left\{ \|v(\cdot, t)\|_{BV}, \|v(\cdot, t) - 1\|_{L_x^1}, \|v(\cdot, t) - 1\|_{L_x^\infty}, \sqrt{t} \|v_t(\cdot, t)\|_{L_x^\infty} \right\} \leq 2C_{\#} \delta, \\ \left| v(\cdot, t) \right|_{x=z^-}^{x=z^+} \leq 2 \left| v_0^*(\cdot) \right|_{x=z^-}^{x=z^+}, \quad z \in \mathcal{D}, \end{array} \right.$$

How to prove uniqueness?

Answer: we need more regularity

Regularity

Wang-Yu-Z.-2022

Suppose the initial data $(v_0^*, u_0^*, \theta_0^*)$ are small. Then we have

- ① There exists a positive constant $C_\#$ such that when $t \in (0, t_\#)$, the solution satisfies

$$\max \left\{ \sqrt{t} \|u_t(\cdot, t)\|_{L_x^1}, t \|u_t(\cdot, t)\|_{L_x^\infty}, \sqrt{t} \|\theta_t(\cdot, t)\|_{L_x^1}, t \|\theta_t(\cdot, t)\|_{L_x^\infty} \right\} \leq 2C_\# \delta. \quad (3.3)$$

- ② The specific volume $v(x, t)$ satisfies the following Hölder continuity in time properties for $0 \leq s < t$,

$$\begin{cases} \|v(\cdot, t) - v(\cdot, s)\|_{BV} \leq O(1)\delta \frac{(t-s)|\log(t-s)|}{\sqrt{t}} \\ \|v(\cdot, t) - v(\cdot, s)\|_{L^\infty} \leq O(1)\delta \frac{t-s}{\sqrt{t}}, \\ \|v(\cdot, t) - v(\cdot, s)\|_{L^1} \leq O(1)\delta(t-s). \end{cases} \quad (3.4)$$

Singularity cancel

Key point:

$$\begin{aligned}
 u_t(x, t) = & \int_{\mathbb{R}} H_t \left(x, t; y, 0; \frac{1}{v} \right) u_0(y) dy + \int_{\mathbb{R} \setminus \mathcal{D}} H_y \left(x, t; y, t; \frac{1}{v} \right) p(v(y, t)) dy \\
 & + \int_0^{\frac{t}{2}} \int_{\mathbb{R} \setminus \mathcal{D}} H_{ty} \left(x, t; y, s; \frac{1}{v} \right) p(v(y, s)) dy ds \\
 & + \int_{\frac{t}{2}}^t \int_{\mathbb{R} \setminus \mathcal{D}} H_{ty} \left(x, t; y, s; \frac{1}{v} \right) (p(v(y, s)) - p(v(y, t))) dy ds \\
 & + \int_{\mathbb{R} \setminus \mathcal{D}} \left(\int_{\frac{t}{2}}^t H_{ty} \left(x, t; y, s; \frac{1}{v} \right) ds \right) p(v(y, t)) dy.
 \end{aligned}$$

Singularities in red terms are canceled!

Regularity in u

Regularity in u .

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$$\begin{cases} \|u_t(\cdot, t)\|_{L^\infty(\mathbb{R})} \leq O(1)\frac{\delta}{t}, & \|u_t(\cdot, t)\|_{L^1(\mathbb{R})} \leq O(1)\frac{\delta}{t}, \\ |u_t(x+h, t) - u_t(x, t)| \\ \leq O(1)\frac{\delta}{t^{3/2}}|h| + O(1)\left(\frac{\delta^2}{t^{3/2}} + \frac{\delta}{t}\right)|h||\log|h|^2 \quad \text{for } |h| < 1. \end{cases}$$

Uniqueness and regularity

$$\left\{ \begin{array}{l} v(x, t) \in C([0, t_{\#}); L^1(\mathbb{R}) \cap L^\infty(\mathbb{R}) \cap BV), \\ u(x, t) \in L^\infty(0, t_{\#}; W^{1,1}(\mathbb{R}) \cap L^\infty(\mathbb{R})), \quad \sqrt{t}u_x(x, t) \in L^\infty(0, t_{\#}; L^\infty(\mathbb{R})), \\ \sqrt{t}u_t(x, t) \in L^\infty(0, t_{\#}; L^1(\mathbb{R})), \quad tu_t(x, t) \in L^\infty(0, t_{\#}; L^\infty(\mathbb{R})), \\ \theta(x, t) \in L^\infty(0, t_{\#}; W^{1,1}(\mathbb{R}) \cap L^\infty(\mathbb{R})), \quad \sqrt{t}\theta_x(x, t) \in L^\infty(0, t_{\#}; L^\infty(\mathbb{R})), \\ \sqrt{t}\theta_t(x, t) \in L^\infty(0, t_{\#}; L^1(\mathbb{R})), \quad t\theta_t(x, t) \in L^\infty(0, t_{\#}; L^\infty(\mathbb{R})). \end{array} \right.$$

The continuity of v implies every solution in this space can be represented by Green's function, then comparison principle estimates imply uniqueness.

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Conservative form

The propagation is governed by hyperbolic part, thus consider the conservative form:

$$\begin{cases} v_t - u_x = 0 \\ u_t + p_x = \left(\frac{\mu u_x}{v} \right)_x \\ \left(e + \frac{1}{2} u^2 \right)_t + (pu)_x = \left(\frac{\kappa}{v} \theta_x + \frac{\mu}{v} u u_x \right)_x, \end{cases} \quad (4.5)$$

where $e = c_v \theta$.

weak solution of non-conservative form is a weak solution to above?

Yes, we have enough regularity!

Linearization around constant states

$$U_t + F(U)_x = (B(U)U_x)_x \iff U_t + F'(U)U_x = (B(U)U_x)_x.$$

$$U = \begin{pmatrix} v \\ u \\ E \end{pmatrix}, \quad F(U) = \begin{pmatrix} -u \\ p \\ pu \end{pmatrix}, \quad F'(U) = \begin{pmatrix} 0 & -1 & 0 \\ p_v & -p_e u & p_e \\ p_v u & p - p_e u^2 & p_e u \end{pmatrix},$$

$$B(U) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \frac{\mu}{v} & 0 \\ 0 & \left(\frac{\mu}{v} - \frac{\kappa \theta_e}{v}\right) u & \frac{\kappa \theta_e}{v} \end{pmatrix}.$$

Long-time Green's function

The Green's function of the linearized system is a matrix value function such that

$$\begin{aligned}\partial_t \mathbb{G}(x, t; \bar{U}) &= (-F'(\bar{U})\partial_x + B(\bar{U})\partial_{xx}) \mathbb{G}(x, t; \bar{U}), \\ \mathbb{G}(x, 0; \bar{U}) &= \delta(x)I,\end{aligned}$$

Denote

$$\mathbb{G} = \begin{pmatrix} \mathbb{G}_{11} & \mathbb{G}_{12} & \mathbb{G}_{13} \\ \mathbb{G}_{21} & \mathbb{G}_{22} & \mathbb{G}_{23} \\ \mathbb{G}_{31} & \mathbb{G}_{32} & \mathbb{G}_{33} \end{pmatrix}. \quad (4.6)$$

Integral transform

$$\begin{aligned}\partial_t \hat{\mathbb{G}}(\eta, t; \bar{U}) &= (-i\eta F'(\bar{U}) - \eta^2 B(\bar{U})) \hat{\mathbb{G}}(\eta, t; \bar{U}), \\ \hat{\mathbb{G}}(\eta, 0; \bar{U}) &= I.\end{aligned}$$

Then, the Green's function is

$$\hat{\mathbb{G}}(\eta, t; \bar{U}) = \mathcal{L}^{-1}[(s + i\eta F'(\bar{U}) + \eta^2 B(\bar{U}))^{-1}],$$

Representation of the Green's function

$$\begin{aligned} & \mathcal{L}^{-1}[(s + i\eta F'(\bar{U}) + \eta^2 B(\bar{U}))^{-1}] \\ &= \sum_{j=1}^3 \text{Res}[e^{st} (s + i\eta F'(\bar{U}) + \eta^2 B(\bar{U}))^{-1}, \lambda_j], \end{aligned}$$

By residue formula, we have

$$\hat{\mathbb{G}}(\eta, t; \bar{U}) = \sum_{j=1}^3 e^{\lambda_j t} \hat{\mathbb{M}}_j, \quad \hat{\mathbb{M}}_j = \frac{\text{adj}(s + i\eta F'(\bar{U}) + \eta^2 B(\bar{U}))|_{s=\lambda_j}}{\prod_{k \neq j} (\lambda_j - \lambda_k)},$$

High frequency

$$\begin{aligned}\lambda_1 &= \frac{\nu p_\nu}{\mu} + \dots \\ \lambda_2 &= -\frac{\eta^2 \mu}{\nu} + \dots \\ \lambda_3 &= -\frac{\eta^2 \kappa \theta_e}{\nu} + \dots\end{aligned}\tag{4.7}$$

Low frequency

$$\lambda_1 = \frac{\eta^2 \kappa \theta_e p_v}{v (p p_e - p_v)} + \dots$$

$$\lambda_2 = -i\eta \sqrt{p p_e - p_v} - \frac{\eta^2 (\kappa p \theta_e p_e + \mu p p_e - \mu p_v)}{2v (p p_e - p_v)} + \dots \quad (4.8)$$

$$\lambda_3 = i\eta \sqrt{p p_e - p_v} - \frac{\eta^2 (\kappa p \theta_e p_e + \mu p p_e - \mu p_v)}{2v (p p_e - p_v)} \dots$$

Green's function

$$\begin{cases} \mathbb{G}^{*,1}(x, t) = e^{\frac{vp_V}{\mu}t} \delta(x) M_1^{*,0} + O(1) e^{-\sigma_0^* t - \sigma_0 |x|}, \\ \mathbb{G}^{*,j}(x, t) = \begin{cases} O(1) e^{-\sigma_0^* t - \sigma_0 |x|} + \frac{e^{\beta_j^* t}}{\sqrt{4\pi\alpha_j^* t}} e^{-\frac{x^2}{4\alpha_j^* t}} M_j^{*,0}, & t < 1, \\ O(1) e^{-\sigma_0^* t - \sigma_0 |x|}, & k = 0, 1, 2, \quad j = 2, 3, \quad t \geq 1 \end{cases} \end{cases}$$

Combine with short-time Green's function

The most singular term is $\mathbb{G}_{22}$ and $\mathbb{G}_{33}$, thus define:

$$\begin{cases} G_{22}(x, t; y, \tau) = \mathcal{X}\left(\frac{t-\tau}{\nu_0}\right) H\left(x, t; y, \tau; \frac{\mu}{\nu}\right) + \left(1 - \mathcal{X}\left(\frac{t-\tau}{\nu_0}\right)\right) \mathbb{G}_{22}(x-y; t-\tau), \\ G_{33}(x, t; y, \tau) = \mathcal{X}\left(\frac{t-\tau}{\nu_0}\right) H\left(x, t; y, \tau; \frac{\kappa}{c_v \nu}\right) + \left(1 - \mathcal{X}\left(\frac{t-\tau}{\nu_0}\right)\right) \mathbb{G}_{33}(x-y; t-\tau). \end{cases} \quad (4.9)$$

Then, redefine

$$\mathbb{G} = \begin{pmatrix} \mathbb{G}_{11} & \mathbb{G}_{12} & \mathbb{G}_{13} \\ \mathbb{G}_{21} & \textcolor{red}{G}_{22} & \mathbb{G}_{23} \\ \mathbb{G}_{31} & \mathbb{G}_{32} & \textcolor{red}{G}_{33} \end{pmatrix}. \quad (4.10)$$

Integral representation

Convolve with original equation to represent (v, u, θ) (for instance):

$$\begin{aligned}
 & u(x, t) \\
 &= \int_{\mathbb{R}} \mathbb{G}_{21}(x - y, t)(v(y, 0) - 1)dy + \int_{\mathbb{R}} G_{22}(x, t; y, 0)u(y, 0)dy \\
 &+ \int_{\mathbb{R}} \mathbb{G}_{23}(x - y, t)(E(y, 0) - c_v)dy + \sum_{i=1}^3 \mathcal{R}_i^u,
 \end{aligned}$$

Global solution

Wang-Yu-Z.-2022

Suppose

$$\begin{aligned} & \|v_0 - 1\|_{BV} + \|u_0\|_{BV} + \|\theta_0 - 1\|_{BV} \\ & + \|v_0 - 1\|_{L^1} + \|u_0\|_{L^1} + \|\theta_0 - 1\|_{L^1} < \delta^*. \end{aligned}$$

Then, the local solution is global existent and has dissipation as below:

$$\begin{aligned} & \|\sqrt{t+1}(v-1)\|_\infty + \|\sqrt{t+1}u\|_\infty + \|\sqrt{t+1}(\theta-1)\|_\infty \\ & + \|\sqrt{t}u_x\|_\infty + \|\sqrt{t}\theta_x\|_\infty < \delta. \end{aligned}$$

Thanks for attention!



Happy Birthday to Professor Shih-Hsien Yu!